

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. DEPLOYMENT ENVIRONMENT :

FitBuddy is engineered as a dynamic web application powered by the FastAPI framework. It does not compile into a standalone desktop executable (. exe). Instead, it is executed locally within a Python Virtual Environment, making it highly portable.

3. SYSTEM PREREQUISITES

- **Base Interpreter:** Python 3 must be installed and added to the system PATH.
- **Network Dependency:** A stable internet connection is required to facilitate communication with the Google Gemini AI endpoint.

4. RUNTIME EXECUTION INSTRUCTIONS

Step 1: Virtual Environment Initialization

Extract the project folder, open your terminal/command prompt, and create a virtual environment to isolate project dependencies:

```
python -m venv env
env\Scripts\activate
```

Step 2: Dependency Installation

Install the core libraries explicitly required for the application to function:

```
pip install fastapi uvicorn sqlalchemy google-generativeai
jinja2 python-multipart
```

Step 3: Environment Configuration The application requires a valid Google API key to generate workout plans. You must set this within your environment variables before running the server:

- **Windows (Command Prompt):** set
GOOGLE_API_KEY="your_actual_api_key_here"
- **Windows (PowerShell):**
\$env:GOOGLE_API_KEY="your_actual_api_key_here"
- **Linux/Mac:** export
GOOGLE_API_KEY="your_actual_api_key_here"

Step 4: Launch the ASGI Web Server Boot up the backend application utilizing the Uvicorn server command:

```
uvicorn app.main:app --reload
```

Step 5: Access the Interface

Open a web browser and navigate to the local host address to begin generating customized fitness plans:

<http://127.0.0.1:8000>

5. RUNTIME TROUBLESHOOTING

- **ModuleNotFoundError:** If the server fails to start, ensure that your virtual environment is actively running and all dependencies (FastAPI, SQLAlchemy, google-generativeai) were successfully installed.
- **Server Timeout/Hanging:** If the API experiences latency, the implemented 10-second timeout will automatically safely abort the request and display a fallback UI error rather than crashing the Uvicorn server.
- **Database Errors:** If SQLite throws locked database errors, stop the server (CTRL+C), delete the local `fitbuddy.db` file to reset the schema, and restart the Uvicorn server.